

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	K.Dhilip
Register Number	192365013

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

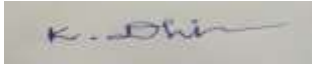
Total Marks: 25

Code of Conduct

I K.Dhilip/192365013 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

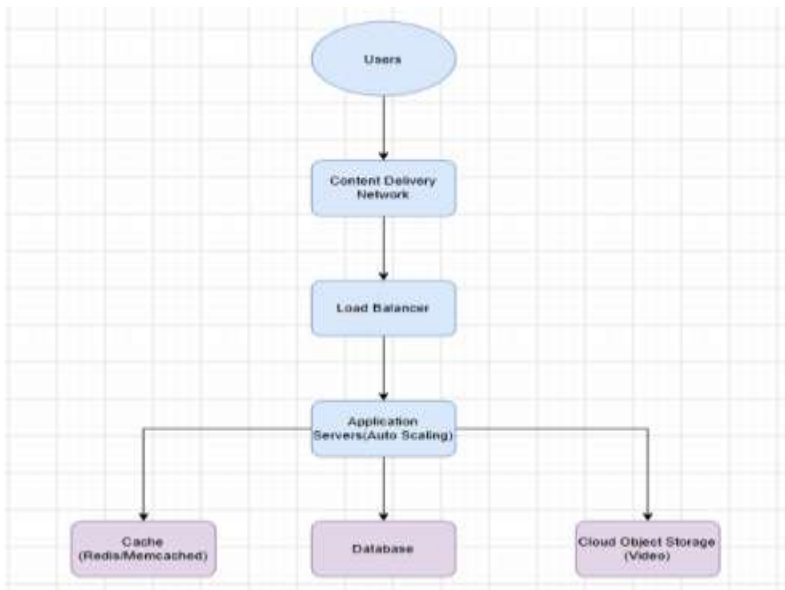
1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Cloud Architecture Using Scalability and Caching Techniques

A media company that streams videos to millions of users needs a cloud architecture that is fast, reliable, and capable of handling sudden increases in traffic. The following architecture can meet these requirements:

- Users access the platform through web browsers, mobile apps, or smart TVs.
- A Content Delivery Network (CDN) stores and delivers video content from servers located close to users. This reduces buffering and improves streaming speed.
- A Load Balancer distributes incoming requests across multiple application servers so that no single server becomes overloaded.
- The Application Servers handle user authentication, recommendations, video requests, and other business logic.
- Auto-Scaling Groups automatically increase or decrease the number of servers based on user demand.
- Frequently accessed information, such as user sessions and video metadata, is stored in an **in-memory** cache (such as Redis or Memcached) for faster access.
- Videos are stored in cloud object storage, while user information and streaming history are maintained in a managed cloud database.
- Monitoring services continuously track system performance and generate alerts if any issue occurs.

Architecture Diagram:



How Load Balancing and Auto-Scaling Ensure Smooth Streaming

Load Balancing

Load balancing plays a key role in improving streaming performance.

- It distributes user requests evenly across multiple servers.
- It prevents any single server from becoming overloaded.
- If one server fails, traffic is automatically redirected to healthy servers.
- It improves system availability and provides uninterrupted video playback.

Example: If 5 million users start watching a live cricket match at the same time, the load balancer shares the requests among many servers, ensuring smooth streaming without delays.

Auto-Scaling

Auto-scaling automatically adjusts the number of servers according to user demand.

- During peak hours, new servers are added automatically.
- During low-traffic periods, unnecessary servers are removed.
- This ensures consistent performance while avoiding wasted resources.
- Users experience minimal buffering even during sudden traffic spikes.

Example: When a popular movie is released, the cloud platform can automatically launch additional servers to handle the increased number of viewers. After demand decreases, the extra servers are shut down.

Cost Optimization Strategies

A streaming platform can reduce operational costs by using the following strategies:

- **Use CDN Caching:** Deliver videos from edge locations to reduce bandwidth usage and lower server load.
- **Auto-Scaling:** Run only the required number of servers instead of keeping many servers active all the time.
- **Object Storage:** Store videos in cost-effective cloud storage instead of expensive high-performance disks.

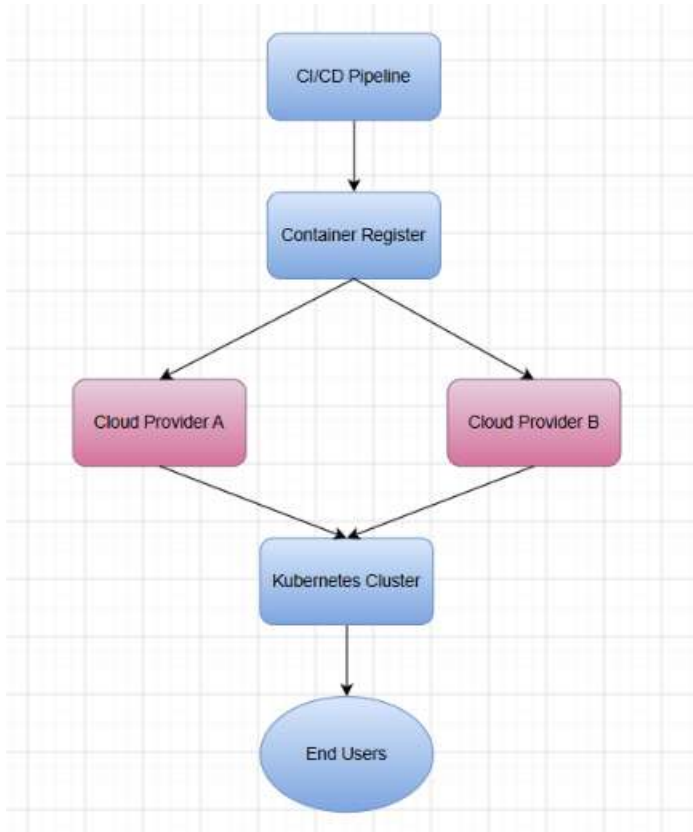
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Solution Using Container Orchestration and Multi-Cloud Strategies

A fintech startup wants to deploy its applications across multiple cloud providers to improve reliability and avoid depending on a single vendor. However, managing containers on different cloud platforms can be difficult. A practical solution is to use container orchestration with a multi-cloud strategy.

- Package each application as a Docker container to ensure it runs the same way on every cloud platform.
- Use Kubernetes as the container orchestration platform to manage, deploy, and monitor containers across multiple cloud providers.
- Store container images in a centralized container registry so every cloud environment uses the same application version.
- Automate deployments using CI/CD pipelines, ensuring that updates are tested and deployed consistently across all cloud environments.
- Use Infrastructure as Code (IaC) tools such as Terraform to create and manage cloud infrastructure in a consistent manner.

Architecture Diagram:



Handling Service Discovery and Scaling Efficiently

Service Discovery

In a containerized environment, containers are created and removed dynamically. Therefore, applications need an easy way to locate each other.

- Kubernetes provides built-in Service Discovery using DNS.
- Every service receives a unique DNS name, allowing containers to communicate without knowing each other's IP addresses.
- If a container fails or is replaced, Kubernetes automatically updates the service information.
- This enables reliable communication between application components.

Example: A payment service can simply contact the authentication service using its service name, even if the authentication containers are restarted or moved to another node.

Scaling

Kubernetes automatically adjusts application capacity based on workload.

- Horizontal Pod Autoscaler (HPA) increases or decreases the number of containers depending on CPU, memory, or custom metrics.
- Cluster Autoscaler adds or removes virtual machines when more computing resources are needed.
- During high transaction periods, additional containers are started automatically.
- When traffic decreases, unused containers are removed to reduce costs.

Example:

During an online shopping festival, payment requests may increase significantly. Kubernetes automatically launches more payment service containers to maintain fast response times.

Benefits and Risks of a Multi-Cloud Deployment Model

Benefits

- **High Availability:** If one cloud provider experiences an outage, applications can continue running on another provider.
- **Avoids Vendor Lock-in:** The company is not dependent on a single cloud vendor.
- **Better Disaster Recovery:** Data and services can be replicated across multiple cloud platforms.

Risks

- **Higher Management Complexity:** Managing multiple cloud environments requires additional expertise.
- **Increased Security Challenges:** Security policies must be implemented consistently across all cloud providers.
- **Data Synchronization Issues:** Keeping data consistent between different clouds can be difficult.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

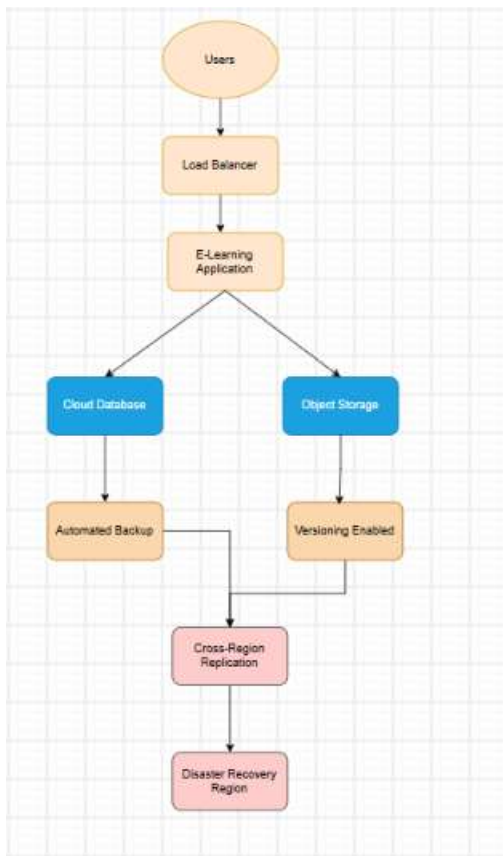
Cloud-Native Backup, Replication, and Versioning Solution

An e-learning platform stores important data such as student records, course materials, videos, assignments, and examination results. If this data is accidentally deleted, the organization needs a reliable backup and disaster recovery (DR) solution to restore services quickly.

A cloud-native solution can include the following components:

- Store all learning content in cloud object storage with versioning enabled. This allows deleted or modified files to be restored easily.
- Schedule automatic backups of databases and application data at regular intervals.
- Enable cross-region replication so that data is copied to another cloud region for disaster recovery.
- Use snapshots of virtual machines and storage volumes to recover entire systems if required.
- Deploy the application across multiple availability zones (AZs) to improve availability.
- Continuously monitor backups and perform regular recovery tests to ensure backup integrity.

Architecture Diagram:



Achieving Recovery Time Objective (RTO) and Recovery Point Objective (RPO)

Recovery Time Objective (RTO)

RTO is the maximum time allowed to restore the application after a failure.

To achieve a low RTO:

- Maintain a standby environment in another cloud region.
- Use automated recovery scripts to restart applications quickly.
- Store infrastructure configurations using Infrastructure as Code (IaC).
- Use load balancers to redirect users to the disaster recovery environment if the primary site fails.

Example: If the target RTO is 30 minutes, the backup system and standby servers should restore the application within 30 minutes after an outage.

Recovery Point Objective (RPO)

RPO is the maximum amount of data that can be lost during a failure.

To achieve a low RPO:

- Perform frequent automated database backups.
- Enable continuous data replication to the disaster recovery site.
- Use storage versioning to recover accidentally deleted files.
- Synchronize databases regularly between the primary and backup regions.

Example: If the RPO is 5 minutes, the organization should lose no more than five minutes of recent data during a disaster.

Steps to Ensure Business Continuity During Failures

The following steps help maintain continuous service even during unexpected failures:

1. **Enable Automatic Backups:** Schedule daily full backups and frequent incremental backups.
2. **Use Cross-Region Replication:** Keep a copy of critical data in another cloud region.
3. **Enable File Versioning:** Recover deleted or modified course materials easily.
4. **Deploy Across Multiple Availability Zones:** Ensure applications remain available even if one data center fails.
5. **Implement Automated Failover:** Redirect user traffic to the disaster recovery environment automatically.
6. **Monitor System Health:** Use cloud monitoring tools to detect failures and send instant alerts.
7. **Perform Regular Disaster Recovery Drills:** Test backup restoration and failover procedures periodically.
8. **Document Recovery Procedures:** Maintain a disaster recovery plan so administrators know exactly how to restore services.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

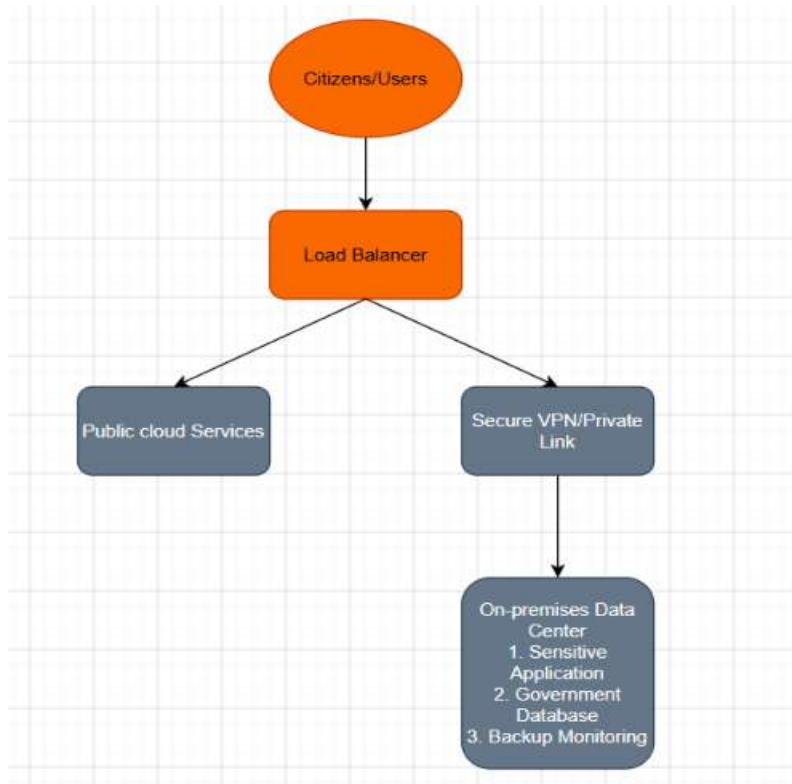
Hybrid Cloud Architecture for Secure Communication

A government agency handles both **sensitive** and **non-sensitive** information. Due to security and legal compliance requirements, sensitive applications and confidential data must remain on-premises, while non-sensitive applications and public services can be hosted in the public cloud.

A secure hybrid cloud architecture can be designed as follows:

- **On-Premises Data Center:** Stores confidential citizen records, financial data, and critical government applications.
- **Public Cloud:** Hosts public websites, online services, portals, and less sensitive applications.
- **Secure VPN or Dedicated Private Connection:** Connects the on-premises environment with the public cloud to ensure secure communication.
- **Firewall and Security Gateway:** Filters incoming and outgoing traffic to prevent unauthorized access.
- **Load Balancer:** Distributes user requests across cloud services for better performance.
- **Monitoring and Logging:** Continuously tracks system activities and detects security threats.

Architecture Diagram:



Identity Management and Encryption

Identity Management

Identity management ensures that only authorized users can access government systems.

- Implement Identity and Access Management (IAM) to control user permissions.
- Use Role-Based Access Control (RBAC) so employees access only the resources required for their jobs.
- Enable Multi-Factor Authentication (MFA) for administrators and government officials.
- Integrate Single Sign-On (SSO) so users can securely access multiple applications using one login.
- Regularly review user accounts and remove access for inactive or former employees.

Example: A finance officer can access financial records, while a public service employee can only access citizen service applications.

Encryption

Encryption protects data from unauthorized access.

- **Encrypt data at rest** using strong encryption standards such as AES-256.
- **Encrypt data in transit** using secure communication protocols such as TLS/SSL.
- Store encryption keys securely using a cloud Key Management Service (KMS) or a hardware security module (HSM).
- Encrypt backups before storing them in cloud storage.
- Rotate encryption keys regularly to improve security.

This ensures that even if data is intercepted or stolen, it cannot be read without the correct encryption keys.

Challenges in Managing Hybrid Cloud Environments

Although hybrid cloud provides flexibility and security, it also introduces several challenges:

- **Complex Management:** Managing both on-premises infrastructure and cloud services requires skilled administrators.
- **Security Risks:** Security policies must be applied consistently across both environments.
- **Compliance Requirements:** Government regulations must be followed for data storage, processing, and auditing.
- **Network Latency:** Communication between on-premises systems and the cloud may introduce delays.
- **Data Synchronization:** Keeping data consistent across multiple environments can be difficult.
- **Monitoring and Troubleshooting:** Tracking performance and identifying issues across different platforms is more complex.
- **Cost Management:** Operating both on-premises infrastructure and cloud resources can increase operational costs.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

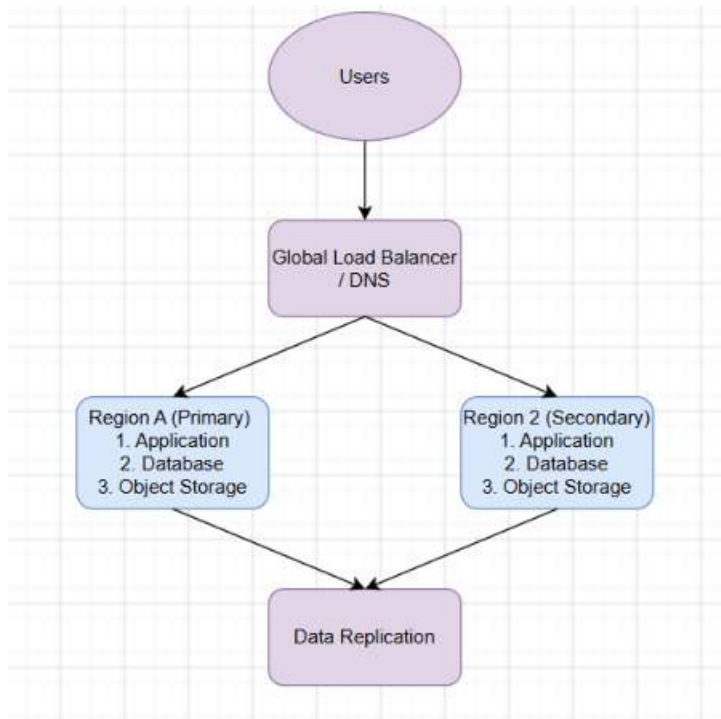
Multi-Region Deployment Strategy

A SaaS (Software as a Service) provider serves users from different locations around the world. To ensure high availability and fault tolerance, the application should continue running even if an entire cloud region becomes unavailable. A multi-region deployment helps achieve this by deploying the application in two or more geographically separate cloud regions.

The proposed strategy includes:

- Deploy the application in multiple cloud regions (Primary and Secondary).
- Place a Global Load Balancer or DNS-based traffic manager in front of the application to direct users to the nearest healthy region.
- Run identical application servers in each region using auto-scaling.
- Store application data in a replicated database that synchronizes data across regions.
- Use shared object storage with cross-region replication for files, images, and backups.
- Maintain a standby environment that can immediately serve users if the primary region fails.

Architecture Diagram:



Data Replication and Failover Mechanisms

Data Replication

Data replication ensures that copies of data are available in multiple regions.

- Use cross-region database replication to synchronize data between the primary and secondary regions.
- Enable object storage replication for application files and backups.
- Perform continuous or near real-time replication to minimize data loss.
- Regularly verify that replicated data remains consistent across regions.

Example: If a customer updates their profile in Region A, the changes are automatically replicated to Region B within seconds.

Failover Mechanisms

Failover automatically switches user traffic to another region during a failure.

- Configure health checks to continuously monitor application availability.
- If the primary region becomes unavailable, the global load balancer redirects traffic to the secondary region.
- Maintain standby application servers so they can immediately handle incoming requests.
- Test failover procedures regularly to ensure smooth recovery during real incidents.

Example:

If Region A experiences a power outage, users are automatically redirected to Region B without noticeable service interruption.

Monitoring and Alerting Strategies

Continuous monitoring helps detect problems before they affect users.

Monitoring

- Monitor CPU, memory, disk usage, and network performance.
- Track application response time and error rates.
- Monitor database performance and replication status.

Alerting

- Configure automatic alerts for high CPU usage, low storage space, or failed servers.
- Send notifications through email, SMS, or collaboration tools when critical issues occur.

Example: If database replication stops or a server becomes unreachable, administrators receive an immediate alert and can take corrective action before users experience major service disruptions.